
Where Does Platonic Convergence Live? Pointwise Curvature of a Galaxy Embedding Manifold and What a Linear Map Can See

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Abstract

1 The Platonic Universe study measured the convergence of foundation-model rep-
2 resentations on galaxy images as one number per pair of embeddings. We ask
3 whether that convergence is concentrated where the embedding manifold is flat.
4 A pointwise curvature field at intrinsic dimension 20 in ambient dimension 768,
5 obtained by differentiating a trained decoder and validated against analytic truth in
6 that regime, finds that mean-curvature magnitude correlates negatively with cross-
7 survey neighbour agreement at every latent dimension, but that local density ex-
8 plains 50 to 80 percent of the association. A second-order expansion of a linear
9 map restricted to the manifold explains why: a readout sees the second fundamen-
10 tal form contracted with its normal component, $\langle w_N, \Pi \rangle$, and an alignment sees the
11 relative second fundamental form of the two embeddings, neither visible to mean-
12 curvature magnitude at large codimension. On a unit-normalized embedding that
13 Hessian splits into a sphere term, proportional to the readout’s own value, and a
14 shape term. On a synthetic surface at production scale the sphere term reads -0.74
15 to -0.80 across samplings that swing the $\|H\|$ partial from -0.29 to $+0.26$; on
16 86,471 labelled galaxies it is negative on every label, the mismatch between the
17 label’s own Hessian and the shape term is negative wherever the label has local
18 structure, and the shape term alone takes the label-dependent sign that mismatch
19 predicts. Curvature-conditioned tests of representational convergence must condi-
20 tion on relative curvature, not on the curvature of one model.

21 1 Introduction

22 The Platonic Representation Hypothesis proposes that large models trained on different data con-
23 verge toward a shared representation. Duraphe et al. [Duraphe et al., 2025] tested it on the sky:
24 galaxies imaged by two surveys (Legacy Survey and Hyper Suprime-Cam, HSC) and embedded by
25 the same vision model share 0.4 to 2 percent of their nearest neighbours, weak but rising with model
26 capacity, one number per pair of embeddings. We ask where on the manifold that agreement lives: if
27 two models disagree most where the embedding surface bends sharply, local mean curvature should
28 associate negatively with local cross-survey agreement. Testing it needs a pointwise curvature instru-
29 ment at intrinsic dimension near 20 in ambient dimension 768, and a design that separates curvature
30 from local density.

31 We contribute (i) a pointwise mean-curvature and second-fundamental-form instrument from a
32 trained decoder, validated against analytic truth in the production regime; (ii) a density-controlled
33 test of the curvature–alignment prediction under two probes, which finds no effect free of density;
34 (iii) the residual expansion of a linear map on a submanifold, which names the curvature such a map
35 responds to and shows mean-curvature magnitude blind to it; and (iv) a known-surface and a real-

d	raw ρ	p	$\rho(\text{dens}, \ H\)$	partial	strat. p	partial, H_{tan}	CKA gap
20	-0.112	$< 10^{-5}$	+0.43	-0.024	0.06-0.07	-0.025	-0.05 to -0.04
25	-0.128	$< 10^{-5}$	+0.32	-0.066	$< 5 \times 10^{-5}$	-0.023	-0.08
32	-0.024	0.009	+0.01	-0.022	0.10-0.17	+0.056	-0.01 to +0.00

Table 1: Curvature versus cross-survey agreement: raw $\rho(\|H\|, \text{MKNN})$; partial residualizing both ranks on k -NN density; p under permutation within density strata; the same partial for $\|H_{\text{tan}}\|$; CKA gap, top minus bottom curvature tertile within density strata (null band $\approx \pm 0.017$).

data test on the linear-readout side, where the expansion’s terms behave as predicted and account for a label-dependent sign.

2 Pointwise curvature from a decoder

Data. DINOv3 ViT-B/16 embeddings of 101,725 galaxies imaged by both HSC and the Legacy Survey [Duraphe et al., 2025], from which we draw 10,000 rows once with a recorded seed. A second set of 86,471 single-survey ViT-B embeddings with catalogue labels (r -band magnitude, photometric redshift, smooth fraction, stellar mass) serves the readout experiment. Every embedding is unit-normalized, so the data lie on S^{767} ; the manifold is intrinsically curved with trivial first homology ($\beta_1 = 0$ under bootstrap bands [Fasy et al., 2014]), so one Euclidean latent chart is appropriate.

Instrument. Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^{768}$ be the decoder of a plain auto-encoder (three hidden layers of width 250, SiLU), fit on 80% of rows; curvature is evaluated on held-out rows. With $J = DF(z)$, metric $g = J^\top J$, normal projector $P_N = I - Jg^{-1}J^\top$ and second fundamental form $\Pi = P_N D^2 F(z)$, the mean-curvature vector is $H = \text{tr}_g \Pi$, by automatic differentiation checked against finite differences. On the unit sphere every d -submanifold carries a radial component of H equal to $-d$ that says nothing about shape, so the map differentiated is $F/\|F\|$ (after which $H_{\text{rad}} = -d$ to 10^{-14}) and $\|H_{\text{tan}}\|$ carries the shape. Decoder second derivatives have served as curvature regularizers [Lee and Park, 2023] and extrinsic curvature of neural manifolds has been read from fitted templates [Acosta et al., 2022]; here the decoder is an instrument, validated against closed-form truth.

Validation. Scored on surfaces with closed-form curvature on four axes: rank (Spearman against true $\|H\|$), direction (cosine of H vectors), magnitude ratio and calibration. At $D = 768$, cubic and ridge surfaces rank 0.53 and 0.97 at $d = 20$ with reconstruction above 0.997 in both, so reconstruction does not predict fidelity; an in-sphere surface $G(z) = Q \text{normalize}([\phi(z); 0.8 h(z); 0 \dots])$ (ϕ the inverse stereographic map $\mathbb{R}^{16} \rightarrow S^{16}$, h four Gaussian bumps, Q a fixed rotation, span 18 dimensions) at $n = 86,471$, $k = 2,048$ gives rank 0.94, direction cosine 0.999, magnitude ratio 1.00, and 0.75 rank under noise at 25% of patch radius. Rank, direction and magnitude come apart at high d and must be reported separately.

3 Curvature and cross-survey convergence

Probe. Per-point mutual k -nearest-neighbour score $\text{MKNN}_i = |N_k(z_1)_i \cap N_k(z_2)_i|/k$ at $k = 20$ between the HSC and Legacy Survey embeddings, the Platonic Universe statistic made local (chance floor 0.002; inference by permutation), correlated with $\|H\|$ over all 10,000 points at $d \in \{20, 25, 32\}$. Centered kernel alignment [Kornblith et al., 2019] with unbiased HSIC, contrasting curvature tertiles within density strata, is the second probe.

Result. The raw association is negative at every d , in the predicted direction, and clears a two-sided permutation threshold (≈ 0.02 ; a planted positive control is detectable to $\rho = 0.02$). But density correlates with agreement at -0.21 at every d and with $\|H\|$ at $d = 20, 25$, so residualizing on density removes 78%, 49% and 8% of the association. Under a stratified null only $d = 25$ survives (-0.066 , replicating across three decoder seeds; CKA agrees at $d = 20$ and 25). Replacing $\|H\|$ by $\|H_{\text{tan}}\|$ collapses the $d = 25$ density partial 2.8-fold into the range of the two d that fail. Convergence is not detectably concentrated by mean-curvature magnitude once density is controlled.

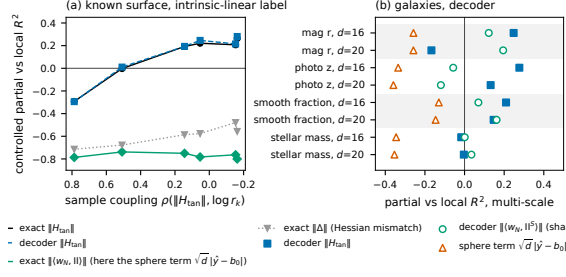


Figure 1: (a) Known surface, intrinsically linear label, exact geometry: three-control partial against local R^2 as the sample’s density–curvature coupling is set by construction (γ from -1 to 1). The decoder tracks the exact $\|H_{\text{tan}}\|$, which follows the sampling; the probe-facing Hessian, here the sphere term, reads -0.74 to -0.80 throughout. (b) Galaxies, 512 anchors, $k = 2,048$: partial under the multi-scale density control for the decoder’s $\|H_{\text{tan}}\|$ (filled squares), the shape term (open circles) and the sphere term (open triangles).

76 4 What a linear map sees on a curved manifold

77 Let $M \subset \mathbb{R}^D$ be d -dimensional; at x_0 take an orthonormal tangent frame J , the second fundamental
 78 form Π (normal-valued, symmetric $d \times d$), $H = \text{tr } \Pi$, and for a vector w its tangent and normal parts
 79 $w_T + w_N$. Writing M locally as $x_0 + Ju + \frac{1}{2}\Pi(u, u) + O(|u|^3)$,

$$\nabla_M(w \cdot x) = J^\top w, \quad \text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle, \quad \Delta_M(w \cdot x) = \langle w, H \rangle. \quad (1)$$

80 A linear function is exactly as non-linear on M as M bends in that function’s normal direction. For
 81 $M \subset S^{D-1}$ the normal bundle splits into the sphere’s radial direction and the rest, $\Pi = \Pi^S - g \hat{x}$,
 82 so

$$\text{Hess}_M(w \cdot x) = \langle w_N, \Pi^S \rangle - (w \cdot \hat{x}) g : \quad (2)$$

83 a linear readout of a unit-normalized embedding is bent by the manifold’s shape and, whatever the
 84 manifold, by the sphere in proportion to the readout’s own value $\hat{y} - b_0$. For a label y on M , a global
 85 affine probe $\hat{y} = w \cdot x + b_0$ and a Gaussian patch with isotropic second moment s^2 in the induced
 86 metric, with $\bar{c}$ the patch-mean residual, $\delta = \nabla_M y - J^\top w$, $\Delta = \text{Hess}_M y - \langle w_N, \Pi \rangle$,

$$\mathbb{E}_{\text{patch}}[(y - \hat{y})^2] = \bar{c}^2 + s^2 \|\delta\|^2 + \frac{1}{2} s^4 \|\Delta\|_F^2 + O(s^4 \|\delta\|) + O(s^6). \quad (3)$$

87 For two embeddings F, G of the same objects and a linear map A aligning their tangent spaces, the
 88 same expansion gives $G - (G_0 + A(F - F_0)) = \frac{1}{2}(\Pi_G - A \Pi_F)(u, u) + o(|u|^2)$: the obstruction
 89 to linear alignment is the *relative* second fundamental form (two rotated copies of one sphere: $\|H\|$
 90 arbitrary, obstruction zero).

91 Three consequences. The curvature a linear map pays for is $\langle w_N, \Pi \rangle$ or $\Pi_G - A \Pi_F$; $\|H\|$ bounds
 92 only a trace ($\|\Pi\|_F^2 \geq \|H\|^2/d$) and need not correlate with either in a 750-dimensional normal
 93 space. The term carries s^4 , so patch radius, hence density, multiplies it. When the label has intrinsic
 94 curvature of its own, Δ carries the residual and the curvature piece can take either sign.

95 5 The probe-facing curvature, on a known surface and on galaxies

96 The readout side can be tested directly: w is known and Π comes from the decoder. Statistic: five-fold
 97 out-of-fold ridge at $\alpha = 100$ from embedding to label; local R^2 inside each of 512 anchors’ 2,048-
 98 neighbourhoods; rank-partial Spearman against local R^2 controlling for log patch radius, local label
 99 variance and count and, as a stronger control, log radius at $k \in \{16, 64, 256, 1024, 2048\}$ (the multi-
 100 scale density control). Per anchor we compute $\|H_{\text{tan}}\|$ and the probe-facing Hessian $\|\langle w_N, \Pi \rangle\|_g$
 101 with $w_N = w - Jg^{-1}J^\top w$, split by Eq. (2) into its shape and sphere terms.

102 **Known surface.** On the validation surface a latent pool three times n is subsampled with weight
 103 $\|H_{\text{tan}}\|^\gamma$, so only where the points sit changes (Figure 1a). The $\|H_{\text{tan}}\|$ partial follows the sampling
 104 from -0.29 at coupling $+0.80$ through zero to a plateau near $+0.22$, and the decoder reproduces

label	$\ H_{\text{tan}}\ $		shape $\ \langle w_N, \Pi^S \rangle\ $		sphere $\sqrt{d} \hat{y} - b_0 $		mismatch $\ \Delta\ $		alignment	
	$d=16$	$d=20$	16	20	16	20	16	20	16	20
mag_r	+0.25	-0.17	+0.12	+0.19	-0.26	-0.26	-0.39	-0.40	+0.35	+0.43
photo_z	+0.28	+0.13	-0.06*	-0.12	-0.33	-0.36	-0.45	-0.44	+0.26	+0.24
smooth_fraction	+0.21	+0.15	+0.07*	+0.16	-0.13	-0.15	-0.09*	-0.17	+0.19	+0.07*
stellar_mass	-0.02*	-0.00*	+0.00*	+0.03*	-0.35	-0.35	-0.04*	-0.09*	+0.01*	+0.02*

Table 2: Galaxies, 512 anchors, $k = 2,048$: rank-partial Spearman against local R^2 under the multi-scale density control; alignment is $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$. * not significant at 0.05 (Freedman–Lane, 2,000 draws); the three-control partials agree in sign in every significant cell.

it. The probe-facing Hessian $\|\langle w_N, \Pi \rangle\|$ reads -0.74 to -0.80 at every sampling. On this surface it is the sphere term of Eq. (2): rank 0.996–1.000 with $\sqrt{d}|\hat{y} - b_0|$, the in-sphere part 20 to 40 times smaller (one in-sphere normal direction) and, where resolvable, following $\|H_{\text{tan}}\|$. The fixture confirms the expansion for the term it can resolve: an intrinsically linear label loses local R^2 where the readout’s own value is large, whatever the sampling. With a nonlinear label the sphere term reads null and $\|\Delta\|$ carries the partial (-0.2 to -0.6), as Eq. (3) predicts.

Galaxies. Here the two terms of Eq. (2) trade places: the shape term dominates the probe-facing Hessian (rank 0.96–0.99 with the full norm; medians 1.7–19 against 0.2–2.6). To read it through Eq. (3) we estimate $\text{Hess}_M y$ per anchor by a least-squares quadratic of the label on the tangent-projected coordinates $u = g^{-1}J^\top(x - x_0)$ of its 2,048 neighbours, where the quadratic coefficient is the covariant Hessian (Π is normal, so the Christoffel term is absorbed). Table 2 shows three regularities. The sphere term is negative on every label and both d (-0.13 to -0.36), as on the fixture. Wherever the label has local structure (magnitude, redshift, smooth fraction; stellar mass is null throughout) the mismatch is negative (-0.09 to -0.45) and the alignment $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$ positive ($+0.07$ to $+0.43$): local R^2 is higher where the manifold bends in the readout’s direction the way the label bends. The shape term alone is therefore label-signed (positive for magnitude and smooth fraction, null to weakly negative for redshift) because it is one of three terms in $\|\Delta\|^2 = \|\text{Hess}_M y\|^2 - 2\langle \text{Hess}_M y, \langle w_N, \Pi \rangle \rangle + \|\langle w_N, \Pi \rangle\|^2$: its sign follows the cross term’s where that is nonzero and the remaining square where it is null. $\|H_{\text{tan}}\|$ changes sign between $d = 16$ and 20 for magnitude: it sits on the curve of Figure 1a (coupling to radius -0.60). Two checks on the instrument: the same regression applied to the readout $w \cdot x$ returns the point cloud’s own $\langle w_N, \Pi \rangle$, whose metric cosine with the decoder’s is 0.55–0.65 at the median; and an independent split-half quadratic-chart estimator of Π gives the same shape-term sign on three of four labels.

6 Discussion

For the Platonic question: convergence on this manifold is not detectably concentrated where mean curvature is low once density is controlled, and the theory says that design could not have found it. MKNN and CKA are invariant to linear maps that leave the relative second fundamental form unchanged, so a per-point convergence score can be conditioned on curvature only through $\Pi_G - A\Pi_F$, which needs both embeddings’ second fundamental forms and a per-point least-squares map between tangent frames: the experiment this paper sets up and does not run. On the readout side, where the map is known, the sphere term and the Hessian mismatch carry the predicted negative sign on the known surface and on galaxies, and the shape term is label-signed because it enters through its alignment with the label’s own Hessian. The lessons transfer to any embedding: mean-curvature magnitude is blind at large codimension, locally scored global maps carry a radius⁴ density interaction, and a unit-normalized embedding bends every linear readout in proportion to its value.

Limitations. One architecture and one survey pair; the validation surface spans 18 of 768 dimensions and cannot resolve the shape term; the label’s Hessian is a noisy local estimate (median R^2 gain over linear 0.10–0.17); the sphere term is not separated from ridge shrinkage of extreme predictions; the relative second fundamental form was not computed; permutation p -values assume anchor exchangeability under overlapping patches.

References

- Kshitij Duraphe, Michael J. Smith, Shashwat Sourav, and John F. Wu. The platonic universe: Do foundation models see the same sky? In *NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences*, 2025. URL <https://arxiv.org/abs/2509.19453>.
- Brittany Terese Fasy, Fabrizio Lecci, Alessandro Rinaldo, Larry Wasserman, Sivaraman Balakrishnan, and Aarti Singh. Confidence sets for persistence diagrams. *Annals of Statistics*, 42(6):2301–2339, 2014.
- Yonghyeon Lee and Frank Chongwoo Park. On explicit curvature regularization in deep generative models. In *2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning (TAG-ML), ICML*, 2023. URL <https://arxiv.org/abs/2309.10237>.
- Francisco Acosta, Sophia Sanborn, Khanh Dao Duc, Manu Madhav, and Nina Miolane. Quantifying extrinsic curvature in neural manifolds. *arXiv preprint*, 2022. URL <https://arxiv.org/abs/2212.10414>.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019.